

NIELIT Virtual Academy

INTERSHIP PROSPECTUS

Name of the Internship: Online Internship in RTL Design using Verilog HDL

Internship Code: IN13

Mode of Conduction: Online

Starting Date: 2nd June 2025

Last date of registration: 1st June 2025

Duration: 6 Weeks

Objective of the Course

- ✓ The internship program aims to enable participants to design reusable Intellectual Property (IP) Cores as building blocks using Verilog HDL.

Outcome of the Course

- ✓ Understand the brief history, present and future, and Design Cycle of VLSI technology.
- ✓ Learn and implement the HDL IPs using Verilog HDL from scratch.
- ✓ Emulate, debug & Characterize reusable IPs
- ✓ Provide hands-on for highly deterministic real-time applications

Course Fee: Rs: 1000/- (inclusive of GST)

Eligibility: Pursuing an Undergraduate level course or above

Methodology

- ✓ Teaching Mode: Self-Pace
- ✓ Access from anywhere anytime
- ✓ Content Access through e-learning portal
- ✓ Doubt Removal Session
- ✓ Covers both Theory & Practical
- ✓ Certification: On completion of the Mini Project

Registration Link: <http://nva.nielit.gov.in>

Contact Details:

- ✓ Course coordinator Name: Ms. Chandralakha R Pillai
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- ✓ Mobile number: 7598730125

Course Structure:

Module No	Module Title
1	VLSI Fundamentals
2	RTL Design Methodology
3	RTL Design Using HDL
4	RTL Simulation and Verification
5	Timing Constraints and Analysis
6	Project

Syllabus:

Detailed Syllabus

Module 1: VLSI fundamentals

- Overview of VLSI technology and its significance
- Introduction to VLSI design flow: Front-end and Back-end
- Fundamentals of MOSFET operation
- CMOS Inverter Characteristics
- CMOS Logic Design
- Transistor Level Schematics and Layouts
- On-Chip Wire Modeling
- Bonding Diagram, Packaging, and Assembly
- Combinational Logic Circuit Critical Path Optimization

Module 2: RTL Design Methodology

- Basics of Register Transfer Level (RTL) design
- Overview of RTL design process and methodology
- Introduction to Hardware Description Languages (HDLs) – Verilog

Module 3: RTL Design Using HDL

- Introduction to Verilog syntax and construct
- Designing combinational and sequential logic using HDL
- Writing RTL code for basic digital circuits

Module 4: RTL Simulation and Verification

- Functional verification techniques for RTL designs
- Introduction to test benches and test bench development
- Simulation and debugging of RTL designs

Module 5: Timing Constraints and Analysis

- Introduction to timing constraints in RTL design
- Timing analysis and optimization techniques
- Setup and hold time violations and resolution

Module 6: Project

- Capstone project
- Project Report Submission